

Show that

$$(\gamma^2)^2 = -I \quad (1)$$

without explicitly choosing a specific representation. Recall $\gamma^0 \equiv \beta$, $\gamma^1 \equiv \beta\alpha_x$, $\gamma^2 \equiv \beta\alpha_y$, and $\gamma^3 \equiv \beta\alpha_z$, and that

$$\begin{aligned} \alpha_x^2 &= \alpha_y^2 = \alpha_z^2 = \beta^2 = I \\ \alpha_j\beta + \beta\alpha_j &= 0 \\ \alpha_i\alpha_k + \alpha_k\alpha_i &= 0 \quad (j \neq k) \end{aligned}$$

Proof. From the definition $\gamma^2 = \beta\alpha_y$, we have

$$\begin{aligned} (\gamma^2)^2 &= (\beta\alpha_y)(\beta\alpha_y) \\ &= \beta\alpha_y\beta\alpha_y. \end{aligned}$$

Since $\alpha_y\beta + \beta\alpha_y = 0$, we have $\alpha_y\beta = -\beta\alpha_y$. Thus,

$$\begin{aligned} (\gamma^2)^2 &= -\beta^2\alpha_y^2 \\ &= -II \\ &= -I. \end{aligned}$$

□